Online Resource for: The center and the ends of the elephant random walk with rare reversals

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Online Resource 1

This Online Resource belongs to the paper "The center and the ends of the elephant random walk with rare reversals". It holds results and proofs that Theorems 1.1 to 1.3 of the paper do not use. Section S1 gives the center and the crossing to second order, Section S2 proves the tail of the law, and Section S3 proves parts (b), (c) and (d) of Proposition 3.25. Numbers with the prefix S refer to this Online Resource, and all other numbers refer to the paper. We keep the notation and the conventions of the paper.

S1 The center and the crossing to second order

Theorem 2.6 gives $\log C_N$ up to an error $O(\varepsilon \log N)$. A lower bound that matches part (b) of that theorem brings the error down to $O(\varepsilon)$.

Proposition S1.1 (the center to second order)**.** *Let $N \geq 64$ be even and $0 < \varepsilon(1+\log N) \leq \frac{1}{100}$. Then*

$$-11\varepsilon - \varepsilon^2(84\log N + 1.05\cdot 10^5) \leq \log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \leq 5200\,\varepsilon.$$

In particular $\left|\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N\right| \leq 5200\,\varepsilon$ there.

In words, C_N is $\frac{4\varepsilon}{N+2}N^{2\varepsilon}$ times a factor $1+O(\varepsilon)$.

Idea of the proof. The proof is in Section S1.A. The upper bound is Theorem 2.6(b). For the lower bound we compare M after the first flip with Pólya's urn conditioned to end at h, which we call the bridge. Then $V_{t+1}(1)/\pi_t$ is the mean over the bridge of a product of factors $1+O(\varepsilon)$, one for each step (Proposition S1.4). By Jensen's inequality the mean of the product is at least the exponential of the mean of the sum of the logarithms. The step at time u adds about $2\varepsilon/u$ to this mean, and the sum over $u \leq N$ is about $2\varepsilon\log N$.

With this bound the proof of Theorem 2.7 gives one more term of the crossing.

Proposition S1.2 (the crossing to second order)**.** *As $N\to\infty$ over even N,*

$$x_N = g_N - \frac{2g_N\log N + g_N^2/2}{N(1+1/g_N)} + O\Big(\frac{\log N}{N}\Big).$$

So $x_N < g_N$ for large N.

In words, the root g_N overshoots the crossing by an amount of order $\log^2 N/N$. The amount is $6\log^2 N/N$ in the limit, but only about 60 percent of that at $N = 51200$ (Table 1).

Proof. We repeat the proof of Theorem 2.7(a) with one more term on each side. The center gains the term $2\varepsilon \log N$ and the end gains the term $-x^2/(2N)$, and both push F up.

Step 1: both sides. Let F be as in the proof of Theorem 2.7, and let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. For large N Proposition S1.1 applies, and its error $5200\,\varepsilon$ is $O(x/N)$, as is the term $\log(1 + 2/N)$ in $\log \frac{4\varepsilon}{N+2}$. For the end we use $\log(1-y) = -y - \frac{y^2}{2} + O(y^3)$ for $0 \le y \le \frac{1}{2}$ with $y = x/N$. So

$$\log C_N = \log x + \log 4 - 2L + \frac{2xL}{N} + O\Big(\frac{x}{N}\Big), \qquad \log E_N = -\log 2 - x - \frac{x^2}{2N} + O\Big(\frac{x}{N}\Big).$$

In the second formula we multiplied the expansion by $N - 1$, and the terms left over are $O(x/N)$ because x^2/N is bounded for $x \le 4L$. Subtracting gives

$$F(x) = G(x) - \Lambda_N + \frac{2xL + x^2/2}{N} + O\Big(\frac{x}{N}\Big) \qquad (1 \le x \le 2\Lambda_N).$$

Step 2: solve for x_N. By Theorem 2.7, x_N lies in this range for large N. So $F(x_N) = 0$ gives $G(x_N) = \Lambda_N - \eta_N$ with $\eta_N = (2x_N L + x_N^2/2)/N + O(L/N)$. Since $G(g_N) = \Lambda_N$, the mean value theorem gives $x_N - g_N = -\eta_N/G'(\xi)$ for some ξ between x_N and g_N. Here $G'(\xi) = 1 + 1/\xi = (1 + 1/g_N)(1 + O(1/N))$, since $|\xi - g_N| = O(L^2/N)$ and $g_N \ge L$ for large N. As $\eta_N = O(L^2/N)$, we get $x_N - g_N = -\eta_N/(1 + 1/g_N) + O(L^2/N^2)$. Replacing x_N by g_N in the main term of η_N changes it by $O(L^3/N^2)$, again because $|x_N - g_N| = O(L^2/N)$. All these errors are $O(L/N)$, which gives the formula. Its main term is of order L^2/N and is larger than the error for large N, so $x_N < g_N$. □

Remark S1.3 (numerical values and open points). The formula of Proposition S1.2 without its error term is within 0.07 of x_N already at $N = 400$ (Table S1). Proposition S1.1 does not identify the term of order ε. At fixed N the center is a polynomial $c_1\varepsilon + c_2\varepsilon^2 + \cdots$ in ε, by Proposition 2.5(b). From its Taylor coefficients, computed in double precision (`verify_elephant_crossing.py`), we get $c_2/c_1 - 2\log N = -1.585878$, -1.546572 and -1.541005 at $N = 1600$, 12800 and 51200. These values approach $2\gamma - 2 - \log 2 = -1.5387\ldots$, but we do not prove this limit here. We do not know whether, for some fixed k, the bound $\log C_N \le \log(c_1\varepsilon) + \varepsilon c_2/c_1 + O(\varepsilon^2(\log N)^k)$ holds uniformly in N in the range of Proposition S1.1. We also do not know an explicit N beyond which $g_N > x_N$.

Table S1: The crossings x_N of Table 1 and their distance to $x_{2\mathrm{nd}}$, the formula of Proposition S1.2 without its error term. These values are numerical and are not used in any proof.

N	x_N	$x_N - x_{2\mathrm{nd}}$
50	3.940487	+0.3452
400	7.629217	+0.0621
1600	10.240565	+0.0179
3200	11.546834	+0.0097
12800	14.158845	+0.0028
51200	16.778468	+0.0008

S1.A The bridge and the lower bound

Fix $1 \le t \le h$, and start the Pólya(c) chain at state 1 at time $t+1$. By Lemma A.1(b), given that it is at h at time N it moves up at time u from b with probability $\bar r_u(b) = \frac{h-b}{N-u}$, whatever c is. So its $h-1$ up steps are placed uniformly at random among the $N-t-1$ steps after time $t+1$. We call this law of the path $(M_u)_{t+1\le u\le N}$ the bridge, and we write $\mathbb{P}_t^{\mathrm{br}}$ and $\mathbb{E}_t^{\mathrm{br}}$ for it. For $1 \le b \le u-1$ let

$$\varpi_u^+(b) = 1 + \varepsilon\frac{u-2b}{b}, \qquad \varpi_u^-(b) = 1 - \varepsilon\frac{u-2b}{u-b},$$

$$\theta_u(b) = \frac{N(u-2b)^2}{2(N-u)\,b(u-b)}, \qquad Q_u(b) = (u-2b)^2\Big[\frac{\bar r_u(b)}{b^2} + \frac{1-\bar r_u(b)}{(u-b)^2}\Big],$$

and let $\varpi_u = \varpi_u^+(M_u)$ if $M_{u+1} = M_u + 1$ and $\varpi_u = \varpi_u^-(M_u)$ otherwise. Put

$$\mathcal{A}_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\,\theta_u(M_u), \qquad \mathcal{B}_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\,Q_u(M_u).$$

Both are nonnegative, and neither depends on ε.

Proposition S1.4 (bridge formula)**.** *Let $0 \le \varepsilon \le 1$ and $1 \le t \le h$. Then*

$$V_{t+1}(1) = \pi_t\,\mathbb{E}_t^{\mathrm{br}}\Big[\prod_{u=t+1}^{N-1}\varpi_u\Big].$$

If $0 < \varepsilon \le \frac12$, then $\log V_{t+1}(1) \ge \log\pi_t + \varepsilon\mathcal{A}_t - \varepsilon^2\mathcal{B}_t$.

Proof. We divide V by the hitting probability of Pólya's urn. The quotient satisfies a recursion whose weights are the step probabilities of the bridge, and each step leaves one factor ϖ_u. For $u \ge t+1$ and $b \in \mathcal{S}_u$ we have $\Phi_u^0(b) > 0$, and we put $R_u(b) = V_u(b)/\Phi_u^0(b)$. By Lemma A.1(a) and (b) with $c = 0$, $\frac{b}{u}\Phi_{u+1}^0(b+1) = \bar r_u(b)\Phi_u^0(b)$ and $(1-\frac{b}{u})\Phi_{u+1}^0(b) = (1-\bar r_u(b))\Phi_u^0(b)$. Dividing $V_u = \mathcal{L}_u V_{u+1}$ by $\Phi_u^0(b)$ gives

$$R_u(b) = (1-\bar r_u(b))\,\frac{1-r_u(b)}{1-b/u}\,R_{u+1}(b) + \bar r_u(b)\,\frac{r_u(b)}{b/u}\,R_{u+1}(b+1),$$

where a term with an infeasible state is absent, and its bridge probability is 0 (if $b = h$ then $\bar r_u(b) = 0$, and if $u - b = h$ then $\bar r_u(b) = 1$). Also $\frac{r_u(b)}{b/u} = \varpi_u^+(b)$ and $\frac{1-r_u(b)}{1-b/u} = \varpi_u^-(b)$. So $R_u(b) = \mathbb{E}_t^{\mathrm{br}}[\varpi_u R_{u+1}(M_{u+1}) \mid M_u = b]$, and $R_N(h) = 1$. Iterating from $u = t+1$ and using $\Phi_{t+1}^0(1) = \pi_t$ gives the formula.

Now let $0 < \varepsilon \le \frac12$. We take the logarithm of the formula and bound it from below in three steps. First, every factor is at least $\frac12$. Indeed if $b \le u/2$ then $\varpi_u^+(b) \ge 1$ and $\varpi_u^-(b) - 1 \in [-\varepsilon, 0]$, and if $b \ge u/2$ the same holds with $+$ and $-$ exchanged. Second, by Jensen's inequality for the logarithm,

$$\log\mathbb{E}_t^{\mathrm{br}}\Big[\prod_u \varpi_u\Big] \ge \sum_u \mathbb{E}_t^{\mathrm{br}}\log\varpi_u \ge \sum_u \mathbb{E}_t^{\mathrm{br}}\big[(\varpi_u - 1) - (\varpi_u - 1)^2\big].$$

The second inequality is $\log(1+y) \ge y - y^2$ for $y \ge -\frac12$, which holds since the difference vanishes at 0 and its derivative $y(1+2y)/(1+y)$ has the sign of y. Third, we compute the two means

given $M_u = b$. Write $\bar{r} = \bar{r}_u(b)$. The bridge moves up with probability $\bar{r}$, and then $\varpi_u - 1 = \varepsilon\frac{u-2b}{b}$. Otherwise $\varpi_u - 1 = -\varepsilon\frac{u-2b}{u-b}$. Since $u\bar{r} - b = \frac{N(u-2b)}{2(N-u)}$, this gives

$$\mathbb{E}_t^{\mathrm{br}}[\varpi_u - 1 \mid M_u = b] = \varepsilon(u-2b)\frac{u\bar{r}-b}{b(u-b)} = \varepsilon\,\theta_u(b), \qquad \mathbb{E}_t^{\mathrm{br}}[(\varpi_u - 1)^2 \mid M_u = b] = \varepsilon^2 Q_u(b).$$

So the sum above is $\varepsilon\mathcal{A}_t - \varepsilon^2\mathcal{B}_t$, which is the bound. □

Lemma S1.5 (the weights w_t)**.** *For each fixed t, $w_t \to t2^{-t-1}$ as $N \to \infty$. For every even $N \geq 2$, $\sum_t w_t(t+1)^3 \leq 124$.*

Proof. Each fraction in $\pi_t = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}$ tends to $\frac{1}{2}$, so $N\pi_t \to t2^{1-t}$, and $Nc_1 \to 4$. This is the first claim. For the second claim we write w_t and $(t+1)^3$ through binomial coefficients in t. By the proof of Lemma 2.4 (with $v = t+1$ and $d = h$), π_t is a constant times $\binom{t}{1}\binom{N-1-t}{h-1}$, and the Chu–Vandermonde identity gives $\sum_t \binom{t}{j}\binom{N-1-t}{h-1} = \binom{N}{h+j}$. With $j = 1$ this gives the constant, namely $w_t = \binom{t}{1}\binom{N-1-t}{h-1}/\binom{N}{h+1}$. Now $t(t+1)^3 = 8\binom{t}{1} + 38\binom{t}{2} + 54\binom{t}{3} + 24\binom{t}{4}$, because both sides are polynomials of degree four that agree at $t = 0, 1, 2, 3, 4$. So $\sum_t w_t(t+1)^3$ is the sum of 8, 38, 54 and 24 times $\binom{N}{h+j}/\binom{N}{h+1}$ for $j = 1, 2, 3, 4$. Since $\binom{N}{h+j} \leq \binom{N}{h+1}$ for $j \geq 1$, it is at most $8 + 38 + 54 + 24 = 124$. □

Under $\mathbb{P}_t^{\mathrm{br}}$ let $K = M_u - 1$ and $K' = m - K$ be the numbers of up steps and of other steps among the $m = u - t - 1$ steps between times $t+1$ and u. So K is hypergeometric, with m draws from $N - t - 1$ steps of which $h - 1$ are up steps. Let $\Omega_u = \{u/8 \leq M_u \leq 7u/8\}$.

Lemma S1.6. *Let $N \geq 16$, $1 \leq t \leq h$ and $t + 1 \leq u \leq N - 1$.*

(a) The second moment is

$$\mathbb{E}_t^{\mathrm{br}}(u - 2M_u)^2 = \frac{(t-1)^2(N-u)^2}{(N-t-1)^2} + \frac{4(u-t-1)(h-1)(h-t)(N-u)}{(N-t-1)^2(N-t-2)} \leq (t-1)^2 + u.$$

(b) If $t \leq \frac{N+2}{4}$, then $\mathcal{A}_t \geq 2(H_{N-1} - H_t - 1 - \frac{1}{t}) \geq 2(\log N - \log t - 3)$.

(c) If $t \leq \frac{N}{8}$ and $u \geq 8(t+1)$, then $\mathbb{P}_t^{\mathrm{br}}(\Omega_u^c) \leq 2e^{-0.0943u}$.

(d) If $t \leq \frac{N}{8}$, then $\mathcal{B}_t \leq 84(t + 1 + \log N) + 683(t+1)^3 + 19100$.

Proof. (a) Here $u - 2M_u = u - 2 - 2K$ and $u - 2 = m + t - 1$. Since $N - 2 = 2(h-1)$, the mean $\mathbb{E}K = \frac{m(h-1)}{N-t-1}$ gives

$$\mathbb{E}(u - 2M_u) = \frac{(m+t-1)(N-t-1) - m(N-2)}{N-t-1} = \frac{(t-1)(N-t-1-m)}{N-t-1} = \frac{(t-1)(N-u)}{N-t-1}.$$

The variance of a hypergeometric variable with m draws from $N - t - 1$ steps, of which $h - 1$ are up steps and $h - t$ are not, is $\operatorname{Var} K = \frac{m(h-1)(h-t)(N-t-1-m)}{(N-t-1)^2(N-t-2)}$, and $N - t - 1 - m = N - u$. So the second term is $4\operatorname{Var} K$, and the identity is $\mathbb{E}(u - 2M_u)^2 = (\mathbb{E}(u - 2M_u))^2 + 4\operatorname{Var} K$. For the bound, the first term is at most $(t-1)^2$ because $N - u \leq N - t - 1$. In the second term $4(h-1)(h-t) = (N-2)(N-2t) = (N-t-1)^2 - (t-1)^2 \leq (N-t-1)^2$, so it is at most $\frac{m(N-u)}{N-t-2}$. This is 0 for $u = t+1$, and for $u \geq t+2$ it is at most u, because $m \leq u$ and $N - u \leq N - t - 2$.

(b) We bound θ_u from below by a multiple of $(u-2b)^2$ and use only the variance term of (a). Since $b(u-b) \le u^2/4$, $\theta_u(b) \ge \frac{2N(u-2b)^2}{(N-u)u^2}$. By (a), $\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^2 \ge 4\operatorname{Var} K$, and in $4\operatorname{Var} K$ the factor $N-u$ cancels. With $4(h-1)(h-t) = (N-2)(N-2t)$ this gives $\mathcal{A}_t \ge \nu_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2}$ with $\nu_t = \frac{2N(N-2)(N-2t)}{(N-t-1)^2(N-t-2)}$. We show that $\nu_t \ge 2$. Here $\nu_1 = \frac{2N}{N-3} > 2$, and for real $t \in [1, \frac{N+2}{4}]$, $\frac{d}{dt}\log\nu_t = -\frac{2}{N-2t} + \frac{2}{N-t-1} + \frac{1}{N-t-2} \ge \frac{3}{N-t-1} - \frac{2}{N-2t} \ge 0$, since the last inequality is equivalent to $4t \le N+2$. Next, the sum is $H_{N-1} - H_t - (t+1)\sum_{u=t+1}^{N-1} u^{-2}$, which is at least $H_{N-1} - H_t - \frac{t+1}{t}$. The sum is also nonnegative. So multiplying by $\nu_t \ge 2$ gives the first inequality, whether this lower bound is positive or not. For the second inequality use $H_{N-1} - H_t \ge \log\frac{N}{t+1} \ge \log N - \log t - \frac{1}{t}$.

(c) Here $m \ge \frac{7}{8}u$, so $\frac{u}{8} \le \frac{m}{7}$. Since $t \ge 1$, $\mathbb{E}K \ge \frac{m}{2}$, and since $t \le \frac{N}{8}$, $\mathbb{E}K' = \frac{m(h-t)}{N-t-1} \ge \frac{3m}{8}$. Hoeffding's inequality also holds for sampling without replacement [1, Section 6], so $\mathbb{P}(K - \mathbb{E}K \le -\sigma m) \le e^{-2\sigma^2 m}$ for $\sigma > 0$, and the same holds for K'. Now $\{M_u < \frac{u}{8}\} \subset \{K - \mathbb{E}K < -\frac{5m}{14}\}$, and since $u - M_u = t + K'$, also $\{u - M_u < \frac{u}{8}\} \subset \{K' - \mathbb{E}K' < -\frac{13m}{56}\}$. So $\mathbb{P}(\Omega_u^c) \le e^{-2(5/14)^2 m} + e^{-2(13/56)^2 m} \le 2e^{-0.10778m} \le 2e^{-0.0943u}$.

(d) We split each term $\mathbb{E}_t^{\mathrm{br}} Q_u(M_u)$ of $\mathcal{B}_t$ into the part on Ω_u and the part off Ω_u, and the three terms of the bound come from the parts on Ω_u, the parts off Ω_u with $u < 8(t+1)$, and the parts off Ω_u with $u \ge 8(t+1)$. In all of them we use that $\bar r(u-b)^2 + (1-\bar r)b^2 \le u^2$, so that $Q_u(b) \le \frac{(u-2b)^2 u^2}{b^2(u-b)^2}$. On Ω_u we have $b(u-b) \ge \frac{7u^2}{64}$, so $Q_u \le \frac{4096}{49}\frac{(u-2b)^2}{u^2} \le 83.6\frac{(u-2b)^2}{u^2}$. By (a) these parts add up to at most $83.6\sum_{u>t}(\frac{(t-1)^2}{u^2} + \frac{1}{u}) \le 83.6(t + H_{N-1}) \le 84(t+1+\log N)$. Off Ω_u we only have $b(u-b) \ge u-1 \ge \frac{u}{2}$, which gives $Q_u \le 4(u-2b)^2 \le 4u^2$. For $u < 8(t+1)$ we bound the probability of Ω_u^c by 1, and these parts add up to at most $\sum_{u<8(t+1)} 4u^2 \le \frac{4}{3}(8(t+1))^3 \le 683(t+1)^3$. For $u \ge 8(t+1)$, (c) bounds the part by $8u^2 e^{-0.0943u}$, and with $q = e^{-0.0943}$ we have $8\sum_{u\ge 1} u^2 q^u = 8q(1+q)/(1-q)^3 < 19100$. □

Proof of Proposition S1.1. The upper bound is Theorem 2.6(b). For the lower bound let $t_0 = \lfloor N/8 \rfloor \ge 8$. By Proposition 2.5(a), $C_N = \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1} V_{t+1}(1)/\pi_t$. This is εc_1 times a mean over the probability vector (w_t). By Jensen's inequality the logarithm of the mean is at least the mean of the logarithms. For $t \le t_0$ we bound $\log(V_{t+1}(1)/\pi_t)$ from below by Proposition S1.4. For $t > t_0$ we use only that it is ≥ 0, which is Lemma A.1(d). This gives

$$\log C_N \ge \log(\varepsilon c_1) + \sum_t w_t (t-1)\log(1-\varepsilon) + \sum_{t\le t_0} w_t\big(\varepsilon\mathcal{A}_t - \varepsilon^2\mathcal{B}_t\big).$$

The first sum. By $\log(1-\varepsilon) \ge -\varepsilon - \varepsilon^2$ and Proposition 2.5(c), it is at least $-(\varepsilon+\varepsilon^2)\sum_t w_t(t-1) \ge -2\varepsilon - 2\varepsilon^2$.

The terms with $\mathcal{A}_t$. Since $t_0 \le \frac{N+2}{4}$, Lemma S1.6(b) gives $\mathcal{A}_t \ge 2(\log N - \log t - 3)$ for $t \le t_0$. Let $w_\le = \sum_{t\le t_0} w_t$. By Jensen's inequality $\sum_t w_t \log t \le \log\sum_t w_t t \le \log 3$, so

$$\sum_{t\le t_0} w_t\,\varepsilon\mathcal{A}_t \ge 2\varepsilon(\log N - 3)w_\le - 2\varepsilon\log 3 \ge 2\varepsilon\log N - 2\varepsilon(1-w_\le)\log N - 6\varepsilon - 2\varepsilon\log 3.$$

By Proposition 2.5(c), $1 - w_\le \le 2\sum_{t>t_0} t2^{-t} = 2(t_0+2)2^{-t_0}$, and $N < 8(t_0+1)$. As $(t_0+2)2^{-t_0}\log(8t_0+8)$ decreases for $t_0 \ge 8$, we get $2\varepsilon(1-w_\le)\log N \le 4\cdot 10\cdot 2^{-8}\log 72\,\varepsilon < 0.67\,\varepsilon$.

The terms with $\mathcal{B}_t$. Here $t \le t_0 \le \frac{N}{8}$, so Lemma S1.6(d) applies. With $\sum_t w_t(t+1) \le 4$ (Proposition 2.5(c)) and Lemma S1.5 it gives $\sum_{t\le t_0} w_t\mathcal{B}_t \le 84(3+1+\log N) + 683\cdot 124 + 19100 = 84\log N + 104128$.

Collecting terms, and using $c_1 = \frac{4}{N+2}$,

$$\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \ge -(2 + 0.67 + 6 + 2\log 3)\,\varepsilon - \varepsilon^2(84\log N + 104130),$$

which gives the lower bound since $2 + 0.67 + 6 + 2\log 3 < 11$. For the last claim, the hypothesis on ε gives $\varepsilon^2(84\log N + 1.05\cdot 10^5) \le \varepsilon\,\frac{84\log N + 1.05\cdot 10^5}{100(1+\log N)}$. The fraction decreases in N, and its value at $N = 64$ is below 205. Since $11 + 205 < 5200$, the two-sided bound follows. □

S2 The tail

We keep the conventions of Section 3 and Appendix B of the paper, and we use the lemmas of that appendix.

This section shows that the law falls like $x/(2m^2)$ beyond the humps.

Theorem S2.1 (the tail)**.** *Let N be even, and let $x = N\varepsilon \ge 1$ depend on N with $x\log^2 N/N \to 0$. Let $\omega_N \to \infty$ with $\omega_N x\log^2(x+1) = o(N)$, and let $\delta'_N \to 0$. Then*

$$f_N(m) = \frac{x}{m(m+1)}\big(1+o(1)\big) \qquad \text{and} \qquad P_N(N-m) = \frac{x}{2m^2}\big(1+o(1)\big),$$

uniformly for $\omega_N x\log^2(x+1) \le m \le \delta'_N N$. We do not compute the absolute constants behind the $o(1)$ terms.

In words, for m well beyond $x\log^2 x$ and well below N, the chance of landing in the bin at distance m from an end is about half the expected number of flipped families with m steps, namely $x/(2m^2)$. The half is there because the first step must point to that end.

Idea of the proof. The proof is below. For the lower bound we use the first flipped family, which has size m with probability about $x/(m(m+1))$ (Lemma B.3), while the other flips change the count only by a relative amount of order $x\log m/m$. For the upper bound we show $\mathbb{P}(M_N \ge m) \le \frac{x}{m}(1 + O(x\log^2 m/m))$, and the unimodality of f_N turns this into a local bound. The mirror term $f_N(N-m)$ is at most $C_N = O(x/N^2)$, which gives the constant $\frac{1}{2}$.

Lemma S2.2 (late start)**.** *Let $j \ge 2$, $t \ge j-1$ and $z \ge 0$. Then $\mathbb{P}(M_t^{[j-1]} \ge z) \le \exp\{-\frac{j-1}{2}(\frac{z}{t} - \varepsilon\log\frac{2t}{j-1})\}$.*

Proof. This is a Chernoff bound with an exponent that changes with time. A chain that starts at 0 at the late time $j-1$ is like an urn that already holds $j-1$ balls of one color, so the fraction $M_k^{[j-1]}/k$ moves slowly. For this reason we take an exponent θ_k of size about $(j-1)/(2k)$.

Write $G_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k \ge j-1$ let $\theta_k = \log\frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. We claim that for $k \ge j-1$

$$\mathbb{E}e^{\theta_{k+1}G_{k+1}} \le e^{\varepsilon\nu/(k+1-\nu)}\,\mathbb{E}e^{\theta_k G_k}.$$

To see this, condition on $G_k = c$. The chain moves up with probability $r_k(c) \le \varepsilon + c/k$, and $1 + y \le e^y$. So

$$\mathbb{E}\big[e^{\theta_{k+1}G_{k+1}} \;\big|\; G_k = c\big] = e^{\theta_{k+1}c}\big(1 + r_k(c)(e^{\theta_{k+1}} - 1)\big) \le \exp\Big\{c\Big(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}\Big) + \frac{\varepsilon\nu}{k+1-\nu}\Big\}.$$

It remains to check that $\theta_{k+1} + \frac{\nu}{k(k+1-\nu)} \le \theta_k$. Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \ge \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. This proves the claim.

We start from $G_{j-1} = 0$ and use the claim for $k = j-1, \dots, t-1$. Since $\sum_{k=j}^{t}\frac{1}{k-\nu} \le \log\frac{t-\nu}{\nu} \le \log\frac{2t}{j-1}$, we get $\mathbb{E}e^{\theta_t G_t} \le \exp\{\varepsilon\nu\log\frac{2t}{j-1}\}$. Finally $\theta_t = -\log(1 - \frac{\nu}{t}) \ge \frac{\nu}{t}$, so Markov's inequality gives $\mathbb{P}(G_t \ge z) \le e^{-\nu z/t}\,\mathbb{E}e^{\theta_t G_t}$, which is the claim. □

Proposition S2.3 (cumulative upper bound)**.** *Assume* (H)*. There is an absolute constant* C *such that for* $6 \le m \le N/2$ *with* $m \ge 16x \log m$ *we have* $\mathbb{P}(M_N \ge m) \le \frac{x}{m}\big(1 + C\,\frac{x \log^2 m}{m}\big)$.

Idea of the proof. Given the first flip at time j, the count M_N is at most $|\mathcal{T}_V| + Z_j$, the size of the first flipped family plus the count from the rest of the tree. The event $|\mathcal{T}_V| \ge m$ gives the main term x/m. The other terms come from a shift of $|\mathcal{T}_V|$ by Z_j, from a large Z_j, and from a late first flip, and each is smaller by a factor of order $x \log^2 m/m$.

Proof. Step 1: split by the first flip. Let $J = \lfloor N/4 \rfloor$ and $Z_j = Z^{(j)}_{N-1}$. First let $2 \le j \le J$. By Lemma 3.3, given $V = j$ the count M_N has the law of $|\mathcal{T}_V| - Y_{|\mathcal{T}_V|} + Z^{(j)}_{N-|\mathcal{T}_V|}$, where, given $V = j$, $|\mathcal{T}_V|$ has the law β_j and is independent of Y and $Z^{(j)}$. This is at most $|\mathcal{T}_V| + Z_j$, because $Y \ge 0$ and $Z^{(j)}$ is nondecreasing in time. So $M_N \ge m$ needs $Z_j > m/8$, or $Z_j \le m/8$ and $|\mathcal{T}_V| \ge m - Z_j$. Hence

$$\mathbb{P}(M_N \ge m \mid V = j) \le \mathbb{P}\Big(Z_j > \frac{m}{8}\Big) + \mathbb{E}\Big[\bar\beta_j(m - Z_j);\, Z_j \le \frac{m}{8}\Big].$$

We compare $\bar\beta_j(m - z)$ with $\bar\beta_j(m)$ for an integer $0 \le z \le m/8$. By (7), each step down from s to $s - 1$ with $s \le m$ multiplies $\bar\beta_j$ by at most $\varrho_j := 1 + \frac{j-1}{N-m-j+2}$, and $\varrho_j \le 1 + \frac{4(j-1)}{N}$ because $N - m - j \ge N/4$. Since $\varrho^z - 1 \le z(\varrho - 1)\varrho^z$ for $\varrho \ge 1$, we get $\bar\beta_j(m - z) \le \bar\beta_j(m)\varrho_j^z \le \bar\beta_j(m)\big(1 + z(\varrho_j - 1)\varrho_j^{m/8}\big)$. So the mean above is at most $\bar\beta_j(m)\big(1 + (\varrho_j - 1)\varrho_j^{m/8}\,\mathbb{E}Z_j\big)$.

On $\{V > J\}$ there is no mark on $2, \dots, J$, so by Lemma 3.1 M_N is a copy of $M_N^{[J]}$ independent of $\{V > J\}$. Altogether

$$\mathbb{P}(M_N \ge m) \le \sum_{j=2}^{J} \mathbb{P}(V = j)\Big[\bar\beta_j(m)\big(1 + (\varrho_j - 1)\varrho_j^{m/8}\,\mathbb{E}Z_j\big) + \mathbb{P}\Big(Z_j > \frac{m}{8}\Big)\Big] + \mathbb{P}(V > J)\,\mathbb{P}\big(M_N^{[J]} \ge m\big).$$

This gives four terms, which we bound in turn. The main term is $\sum_j \mathbb{P}(V = j)\bar\beta_j(m) \le \mathbb{P}(|\mathcal{T}_V| \ge m) \le \frac{x}{m}$ by Lemma B.3.

Step 2: the shift term. This is the sum over $2 \le j \le J$ of $\mathbb{P}(V = j)\,\bar\beta_j(m)\,(\varrho_j - 1)\,\varrho_j^{m/8}\,\mathbb{E}Z_j$. We bound the five factors by

$$\mathbb{P}(V = j) \le \varepsilon, \quad \bar\beta_j(m) \le e^{-\frac{(j-1)(m-1)}{N-1}}, \quad \varrho_j - 1 \le \frac{4(j-1)}{N}, \quad \varrho_j^{m/8} \le e^{\frac{(j-1)m}{2N}}, \quad \mathbb{E}Z_j \le x \log \frac{N}{j-1}.$$

The second is (6), the fourth follows from the third and $1 + y \le e^y$, and the last is Lemma B.2. The two exponentials combine. Since $m \ge 6$ we have $\frac{m-1}{N-1} - \frac{m}{2N} \ge \frac{m-2}{2N} \ge \frac{m}{3N}$. So with $a = \frac{m}{3N}$ and $i = j - 1$ the shift term is at most

$$\frac{4\varepsilon x}{N} \sum_{1 \le i < N/4} i\, e^{-ai} \log \frac{N}{i}.$$

Here $i \le J - 1 < N/4$, so every term is nonnegative. In this sum only the first flips up to a time of order $1/a = 3N/m$ matter, because a later first flip rarely has a family of size m. To bound the sum we write $\log \frac{N}{i} = \log \frac{m}{3} + \log \frac{1}{ai}$. The first part is nonnegative, and $\sum_{i \ge 1} i e^{-ai} \le a^{-2}$. The second part is positive only for $i < 1/a$. There $i \log \frac{1}{ai} \le \frac{1}{ea}$, so these at most $1/a$ terms add up to at most $\frac{1}{ea^2}$. Hence the shift term is at most

$$\frac{4\varepsilon x}{N} \cdot \frac{\log m + 1}{a^2} = \frac{36x^2(\log m + 1)}{m^2} \le \frac{x}{m} \cdot \frac{Cx \log m}{m}.$$

Step 3: the big-Z term. This is $\sum_{j=2}^{J}\mathbb{P}(V=j)\mathbb{P}(Z_j>m/8)$, and again $\mathbb{P}(V=j)\le\varepsilon$. Recall that Z_j counts the disagreeing steps of a walk whose first $j-1$ steps agree. For small j we only have the general tail bound of Lemma B.2, which is of order $x\log m/m$. For large j the late start makes $Z_j\ge m/8$ exponentially unlikely (Lemma S2.2). We split at $j_{\mathcal{K}}=\lfloor\mathcal{K}N/m\rfloor+1$ with $\mathcal{K}=32\log(m/x)$. Note that $m\ge 16x\log 6>28x$, so $\mathcal{K}\ge 2$.

Let $2\le j\le j_{\mathcal{K}}$. Lemma B.2 at the integer $\lfloor m/8\rfloor+1$ gives $\mathbb{P}(Z_j>m/8)\le 8xH_m/m$. There are at most $\mathcal{K}N/m$ such j, and $\mathcal{K}\le 32\log m$. So they contribute at most

$$\varepsilon\cdot\frac{\mathcal{K}N}{m}\cdot\frac{8xH_m}{m}=\frac{x}{m}\cdot\frac{8\mathcal{K}xH_m}{m}\le\frac{x}{m}\cdot\frac{Cx\log^2 m}{m}.$$

Let $j>j_{\mathcal{K}}$, so that $j-1>\mathcal{K}N/m$. We use Lemma S2.2 with $t=N-1$ and the level $m/8$. Its exponent contains the term $\varepsilon\log\frac{2(N-1)}{j-1}$. Here $\frac{2(N-1)}{j-1}\le\frac{2m}{\mathcal{K}}\le m$, so this term is at most $\frac{x\log m}{N}\le\frac{m}{16N}$, by the hypothesis $m\ge 16x\log m$. The other term in the exponent is $\frac{m}{8(N-1)}\ge\frac{m}{8N}$. So the lemma gives $\mathbb{P}(Z_j\ge m/8)\le e^{-(j-1)m/(32N)}$, which is a geometric sequence in j. Summing over $j-1\ge j_{\mathcal{K}}>\mathcal{K}N/m$, these j contribute at most

$$\varepsilon\,\frac{e^{-\mathcal{K}/32}}{1-e^{-m/(32N)}}\le\varepsilon\cdot\frac{x}{m}\Big(\frac{32N}{m}+1\Big)\le\frac{x}{m}\cdot\frac{33x}{m}.$$

Here $e^{-\mathcal{K}/32}=x/m$ by the choice of $\mathcal{K}$, and we used $\frac{1}{1-e^{-y}}\le\frac{1}{y}+1$ for $y>0$ and $\varepsilon\le x/m$.

Step 4: no early flip. If there is no flip up to time J, the walk agrees with X_1 for about $N/4$ steps, and Lemma S2.2 makes $M_N\ge m$ exponentially unlikely in m. Namely $\mathbb{P}(V>J)=(1-\varepsilon)^{J-1}\le e^{\varepsilon-3x/16}$, since $J\ge 3N/16$, and Lemma S2.2 with $j-1=J\in[\frac{3N}{16},\frac{N}{4}]$ and $t=N$ gives $\mathbb{P}(M_N^{[J]}\ge m)\le\exp\{-\frac{3m}{32}+\frac{x}{8}\log\frac{32}{3}\}$. As $x\le m/28$ and $\varepsilon\le\frac14$, the product of the two is at most $e^{1/4-m/13}$, and this is at most $\frac{x}{m}\cdot\frac{Cx\log^2 m}{m}$ because $x\ge 1$. □

Proposition S2.4 (local upper bound)**.** *Assume* (H)*. There are absolute constants A_1 and C such that if $\varepsilon\log N\le\bar{c}$ and $A_1x\log^2(x+1)\le m\le N/8$, then*

$$f_N(m)\le\frac{x}{m(m+1)}\big(1+C\Delta(m)^{1/2}\big),\qquad\Delta(m)=\frac{x\log^2 m}{m}+\varepsilon\log N+\frac{m}{N}.$$

Idea of the proof. To the right of its mode f_N is nonincreasing, so $(k+1)f_N(m)$ is at most the probability of $\{m-k\le M_N\le m\}$, which is a difference of two tails. We bound the first tail from above by Proposition S2.3 and the second from below by summing Proposition B.4. Then we choose k to balance the errors.

Proof. We take $A_1\ge 2A_0$ so large that Lemma B.5 applies and every $m'\ge m/2$ satisfies $m'\ge 6$ and $m'\ge 16x\log m'$ (as $t/\log t$ increases for $t\ge e$).

Step 1: a difference of tails. Let $1\le k\le m/2$. Then $m-k\ge m/2\ge m_h$, and Lemma B.1 with $a=m-k$ and $b=m$ gives $(k+1)f_N(m)\le\mathbb{P}(M_N\ge m-k)-\mathbb{P}(M_N\ge m+1)$.

Step 2: bound each tail. Proposition S2.3 at $m-k$ bounds the first term by $\frac{x}{m-k}+\frac{Cx^2\log^2 m}{m^2}$. For the second we sum Proposition B.4 over $m<\ell\le N/4$. Since $\sum_{m<\ell\le N/4}\frac{1}{\ell(\ell+1)}\ge\frac{1}{m+1}-\frac{4}{N}$, $\frac{\log(\ell+1)}{\ell}$ and $\ell e^{-\ell/9}$ decrease for $\ell\ge 9$, and $me^{-m/9}\le C/m\le C\Delta(m)$, we get $\mathbb{P}(M_N\ge m+1)\ge\frac{x}{m+1}(1-C\Delta(m))$. As $\frac{x}{m-k}-\frac{x}{m+1}=\frac{x(k+1)}{(m-k)(m+1)}$, this gives $f_N(m)\le\frac{x}{(m-k)(m+1)}+\frac{Cx\Delta(m)}{(k+1)m}$.

Step 3: choose k. If $\Delta(m)\le\frac{1}{16}$, take $k=\lceil m\Delta(m)^{1/2}\rceil\le\frac{m}{2}$. Then $\frac{m}{m-k}\le 1+\frac{2k}{m}\le 1+2\Delta^{1/2}+\frac{2}{m}$ and $\frac{\Delta}{k+1}\le\frac{\Delta^{1/2}}{m}$, which gives the claim, as $\frac{1}{m}\le\Delta(m)\le\Delta(m)^{1/2}$. If $\Delta(m)>\frac{1}{16}$,

take $k = \lfloor m/2 \rfloor$. Then $f_N(m) \le \frac{x}{m(m+1)}(2 + C\Delta(m))$, and the claim follows after enlarging C, since $\Delta(m)$ is bounded by an absolute constant ($\frac{x \log^2 m}{m}$ decreases for $m \ge e^2$ and is bounded at $m = A_1 x \log^2(x+1)$). □

Proof of Theorem S2.1. We check that Propositions B.4 and S2.4 apply on the whole range of m and that their error terms tend to 0 uniformly there. Then we show that the mirror term $f_N(N-m)$ is negligible.

Step 1: the hypotheses hold. Put $\ell_x = \log(x+1) \ge \log 2$ and let $\mathcal{R}_N$ be the range $\omega_N x \ell_x^2 \le m \le \delta'_N N$. Since $x \log^2 N/N \to 0$, for large N we have $\varepsilon(1 + \log N) \le \min(\bar{c}, \frac{1}{100})$, $\omega_N \ge \max(A_1, 16)$, $\delta'_N \le \frac{1}{8}$ and $A_0 x \log(x+2) \le N/4$. So (H) holds as well.

Step 2: the errors are small on $\mathcal{R}_N$. The function $\frac{x \log^2 m}{m}$ decreases for $m \ge e^2$. At $m = \omega_N x \ell_x^2$ we have $m \ge e^2$ and $\log m \le \log \omega_N + 3\ell_x$, as $\log x$ and $\log \ell_x$ are at most ℓ_x. So on $\mathcal{R}_N$, uniformly in $x \ge 1$, $\frac{x \log^2 m}{m} \le \frac{(\log \omega_N + 3\ell_x)^2}{\omega_N \ell_x^2} \le \frac{1}{\omega_N}(\frac{\log \omega_N}{\log 2} + 3)^2 \to 0$. Also $m \ge \omega_N \log^2 2 \to \infty$ on $\mathcal{R}_N$. Since $\log(m+1) \le 2 \log^2 m$ for $m \ge 3$, Proposition B.4 has $e_1(m) \le C(\frac{x \log^2 m}{m} + \varepsilon \log N + m e^{-m/9}) \to 0$ uniformly on $\mathcal{R}_N$. Also $\Delta(m) \le \frac{x \log^2 m}{m} + \varepsilon \log N + \delta'_N \to 0$ there, and Proposition S2.4 applies. So $f_N(m) = \frac{x}{m(m+1)}(1 + o(1))$ uniformly on $\mathcal{R}_N$.

Step 3: the mirror term. Lemma B.5 gives $m_h \le A_0 x \ell_x^2 \le h$ for large N, since $x \ell_x^2 \le x \log^2(N+1) = o(N)$. As $N - m \ge h$, Lemma 3.4 gives $0 \le f_N(N-m) \le f_N(h) = C_N$. By Theorem 2.6(b) and $\varepsilon \log N \to 0$, $C_N \le \frac{4x}{N(N+2)} N^{2\varepsilon} e^{5200\varepsilon} = \frac{4x}{N^2}(1 + o(1))$. On $\mathcal{R}_N$ this is at most $\frac{x}{m(m+1)} \cdot 8\delta'^2_N (1 + o(1))$, since $m(m+1) \le 2\delta'^2_N N^2$. Hence $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)] = \frac{x}{2m(m+1)}(1 + o(1))$ uniformly on $\mathcal{R}_N$, and $m(m+1) = m^2(1 + o(1))$ because $m \to \infty$. □

S3 Odd N

We prove parts (b), (c) and (d) of Proposition 3.25. As there, $N \ge 3$ is odd, $c = 1 - 2\varepsilon$ and $\widetilde{C}_N$ is the probability of each of the two central bins.

Proof of Proposition 3.25(b), (c) and (d). (b) We follow the proof of Theorem 2.7(a), with $\widetilde{F}(x) = \log \widetilde{C}_N - \log E_N$ at $\varepsilon = x/N$ in place of F. Let $1 \le x \le 2\Lambda_N$. For large N, Theorem 2.6 applies to the even number $N+1$ at this ε and gives $\log C_{N+1} = \log \frac{4\varepsilon}{N+3} + O(\varepsilon L)$. By (a), $\log \widetilde{C}_N = \log C_{N+1} + O(1/N)$. Here $\log \frac{4\varepsilon}{N+3} = \log x + \log 4 - 2L + O(1/N)$, and $1/N \le xL/N$. So

$$\log \widetilde{C}_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big),$$

which is the estimate for $\log C_N$ in Step 1 of that proof. The formula $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$ holds for every N (Lemma 2.2). So (∗) holds for $\widetilde{F}$, and Step 2 of that proof gives $|x_N - g_N| = O(L^2/N)$ and the claim. Part (b) of Theorem 2.7 follows from part (a) as before.

(c) We argue as in the proof of Proposition 3.7. Let $N \ge 1689$ be odd and take $\varepsilon = 2/N$. Then $\varepsilon(1 + \log(N+1)) \le \frac{1}{100}$, so Theorem 2.6(b) for $N+1$ and part (a) give

$$\widetilde{C}_N \le \frac{N}{N-1} C_{N+1} \le \frac{N}{N-1} \cdot \frac{8}{N(N+3)} (N+1)^{4/N} e^{10400/N}.$$

The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1689$. Also $E_N = \frac{1}{2}(1 - \frac{2}{N})^{N-1} > 0.06$, as in that proof. So $\widetilde{C}_N < E_N$ at $\varepsilon = 2/N$, and hence $\varepsilon_N > 2/N$ and $x_N > 2 \ge 2(1 - \varepsilon_N)$.

Theorem 3.5 then gives $P_N(N-1) > E_N$ at the crossing, which is $P_N(0) < P_N(1)$ by symmetry. This is the hypothesis of the second case of Corollary 3.20.

(d) The proof of Theorem 3.21 uses that N is even in two places only. The first is $x \to \infty$ and $x = O(\log N)$, which holds for odd N by (b). The second is Corollary 3.20 at the end of Step 4, which applies by (c). The rest of that proof uses Theorem 3.9, Lemma 3.10 and $P_N(m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$, and these hold for every N. □

References

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